

***Monalonion dissimulatum* (Miridae) in Peruvian cacao: Integration of DNA sequences and morphology for its identification and the study of its biological cycle in the Amazonas region.**

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ABSTRACT

Monalonion dissimulatum has become an important pest of *Theobroma cacao* in Peru. However, studies focused on its taxonomy and ecology remain scarce and insufficient for the development of effective control strategies. In this context, the present study aimed to evaluate the incidence, characterize the morphology, confirm the molecular identity, and describe the biological cycle of this pest. Pearson correlation and linear regression analyses revealed that incidence was positively and significantly associated with altitude. Morphometric characterization allowed a preliminary identification of *M. dissimulatum*, although sexual dimorphism was not evident. Phylogenetic analysis based on the mitochondrial COI gene confirmed the species-level delimitation of the analyzed samples. The biological cycle comprises seven developmental stages, with a total duration ranging from 34 to 46 days. These findings provide fundamental information on the biology and taxonomy of *M. dissimulatum*, facilitating the development of integrated management strategies.

Keywords: biological cycle, damage, identification, incidence, *Monalonion dissimulatum*, *Theobroma cacao*.

1 INTRODUCTION

The impact of pests and diseases on agricultural production systems transcends economic losses, representing one of the principal threats to global food security and crop sustainability (Donatelli et al. 2017). In tropical and subtropical regions, this impact is intensified by environmental conditions that favor the proliferation of phytophagous organisms (Nega 2025). Within this group, insects of the genus *Monalonion* (Hemiptera: Miridae) are reported as key pests in several crops of economic importance in the Neotropical region (Góngora et al. 2020).

In the Americas, cacao cultivation (*Theobroma cacao* L.) holds considerable social, economic, and cultural significance, providing livelihoods for thousands of rural families (Leiva et al. 2022; Santillan-Culquimboz et al. 2025). However, its production is constrained by various phytosanitary factors, among which infestations of *Monalonion dissimulatum* are particularly notable. Populations of this species have been documented to cause both direct and indirect damage that compromise yield and the quality of the final product (Gamboa et al. 2024).

Direct damage results from its piercing-sucking feeding behavior; by perforating the fruits and extracting sap from the endocarp, the insect induces necrotic lesions that lead to abortion and deformation of young fruits, as well as a reduction in pod size (Vargas Serna et al. 2020). Furthermore, indirect damage is manifested through its role as a primary vector of frosty pod rot (*Moniliophthora roreri*), one of the most devastating diseases affecting cacao, which can cause total losses in severely affected plantations (Díaz-Valderrama et al. 2020; Leiva et al. 2022; Santillan-Culquimboz et al. 2025).

The aggressiveness of *Monalonion* is associated with its capacity to feed throughout its entire biological cycle. This insect exhibits paurometabolous development, with an egg

stage and five nymphal instars before reaching the adult stage, completing its life cycle in approximately 35–45 days (Gamboa et al. 2020; Vargas Serna et al. 2020). Nymphs are apterous, bright orange in color, and highly mobile on the host plant, whereas the elongated eggs, provided with two aeropyles, are inserted endophytically into tender plant tissues (Gamboa et al. 2024). Adults, measuring 10–12 mm in length, display a black head and a dark brown body, with crepuscular or nocturnal habits, feeding on shoots, inflorescences, and young fruits (Vargas Serna et al. 2020; Gamboa et al. 2024). Despite the growing concern regarding the impact of *Monalonion dissimulatum* on cacao production systems, studies specifically focused on its identification and biology associated with this crop remain limited. Although general aspects of the genus *Monalonion* have been documented in other crops of economic relevance, cacao has received comparatively less attention from taxonomic and ecological perspectives (Gamboa et al. 2020, 2024). The available information on the morphological characteristics, biological cycle, and ecology of this pest remains insufficient to support the development of integrated management strategies that are both efficient and specifically adapted to cacao cultivation (Babin 2018; Gamboa et al. 2024). Traditionally, the identification of this species has relied on morphological characters (Gamboa et al. 2020), which has limited its precise taxonomic delimitation due to phenotypic variability influenced by both endogenous and exogenous factors (Karthika et al. 2016). To overcome these limitations, molecular identification based on the mitochondrial marker *cytochrome c oxidase subunit I* (COI) has become established as a robust and standard tool for species delimitation in insects (Jung and Lee 2012; Kim and Jung 2018). Traditionally, the identification of this species has relied on morphological characters (Gamboa et al., 2020), which has limited its precise taxonomic delimitation due to phenotypic variability influenced by both endogenous and exogenous factors (Karthika et al., 2016). To overcome these limitations, molecular identification based on the mitochondrial marker *cytochrome c oxidase subunit I* (COI) has become established as a robust and standard tool for species delimitation in insects (Jung and Lee, 2012; Kim and Jung, 2018). Consequently, the objective of the present study is to comprehensively characterize *Monalonion* in cacao (*Theobroma cacao* L.) in the Amazonas region of Peru through the evaluation of its incidence and associated damage, the confirmation of its identity using morphological characters and molecular markers (COI), and the determination of its biological cycle.

2 METHODOLOGY

2.1 Study Area

The study was conducted in 25 productive cacao (*Theobroma cacao* L.) plots, each approximately 1.0 ha in size, located in the provinces of Bagua and Utcubamba, Amazonas region, Peru, distributed along an altitudinal gradient ranging from 430 to 1155 m above sea level (**Figure 1**). The assessment of incidence and associated damage

covered all plots located in the districts of Cajaruro, El Milagro, Aramango, Imaza, Bagua, Copallín, and La Peca. The biological cycle was studied *in situ* in a plot located in the district of Cajaruro, while specimens for morphological and molecular identification were collected from three farms in the districts of Cajaruro and La Peca. The collected samples were processed at the Plant Health Laboratory of the Universidad Nacional Toribio Rodríguez de Mendoza de Amazonas.

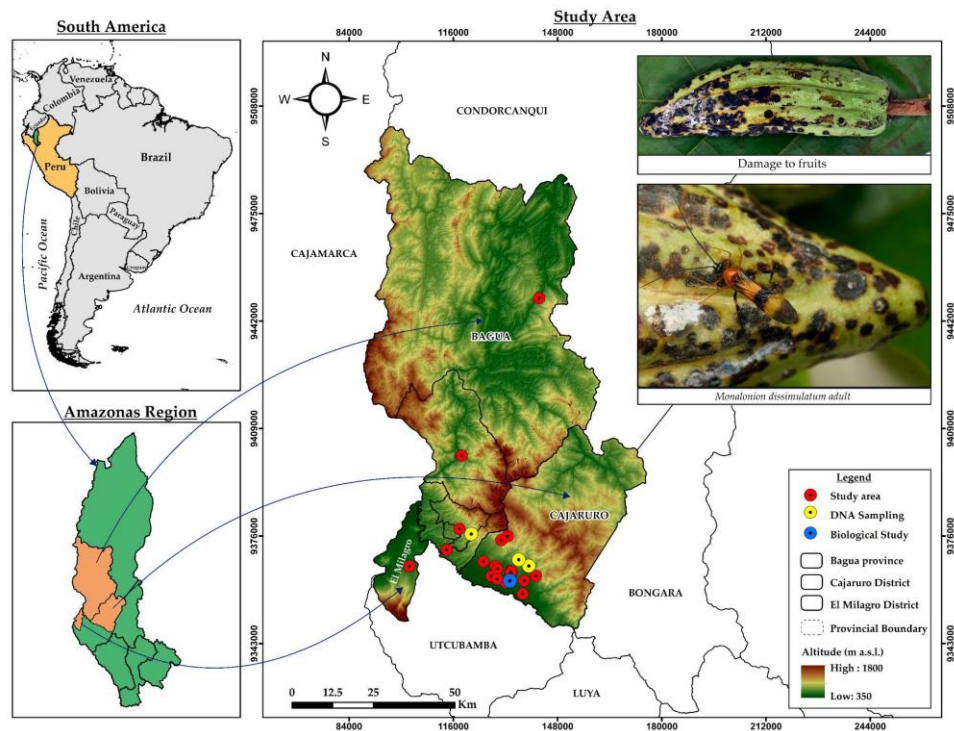


Figure 1: Spatial distribution of the sampling sites of *Monalonion dissimulatum* in cacao plantations in the provinces of Bagua and Utcubamba, Amazonas, Peru (430–1155 m a.s.l.).

2.2 Incidence and Damage Assessment

Incidence was estimated by evaluating the presence or absence of *Monalonion* on cacao plants. In each plot, 286 plants were assessed, a sample size calculated using the finite population formula (Cochran, 1977), considering a planting density of 3×3 m ($1111 \text{ plants ha}^{-1}$). For each plant, a binary variable was recorded, assigning 1 to the presence and 0 to the absence of the pest. Based on these records, the percentage of incidence per plot was calculated.

116 Damage to cacao pods was characterized through direct field observations. The
117 morphological characteristics of the lesions were recorded, including their shape,
118 distribution, and appearance on the fruit surface. In addition, changes in tissue
119 coloration and the presence of necrosis associated with the insect feeding sites were
120 documented.

121 **2.3 Morphological Identification**

122 Specimens of *Monalonion* were collected at all developmental stages. Eggs were
123 extracted from cacao pods through careful dissection using a sterile scalpel. Nymphs
124 were manually collected from infested fruits and branches. Adults were captured
125 through manual collection and with the use of an entomological net. Subsequently, the
126 specimens were transported to the Plant Health Laboratory in ventilated plastic
127 containers containing plant substrate to preserve their viability. Each biological stage
128 was maintained in separate containers.

129 Morphometric documentation was performed using a Nikon SMZ1270
130 stereomicroscope equipped with a Nikon D5 camera. Additionally, high-resolution
131 micrographs were obtained through scanning electron microscopy (SEM), and
132 photographic records were captured with a Sony Alpha IV camera. A total of 45
133 individuals per developmental stage were randomly selected for the measurement of
134 morphometric characters, following the taxonomic descriptors reported for the genus
135 *Monalonion* (Giraldo-Jaramillo et al. 2010; Gamboa et al. 2020).

136 **2.4 DNA Extraction, Sequencing, and Phylogenetic Analysis**

137 Total genomic DNA was extracted from insect samples (thorax) using the Quick-DNA™
138 Tissue/Insect Miniprep Kit (Zymo Research, USA), following the manufacturer's
139 instructions. DNA concentration was quantified by spectrophotometry using an
140 Eppendorf BioSpectrometer® kinetic (Germany) and stored at -20 °C until subsequent
141 amplification by polymerase chain reaction (PCR). Molecular analyses were applied
142 uniformly to the *Monalonion* specimens. Partial sequences of the *cytochrome c oxidase*
143 *subunit I* (COI) gene were amplified using the universal primer pair HCO2198/LCO1490
144 (Folmer et al. 1994).

145 COI gene sequences were obtained by Sanger sequencing and manually edited using
146 Sequencher v5.4.6 (Gene Codes Corporation, Ann Arbor, MI, USA). Consensus sequences
147 were aligned using MEGA X (*Molecular Evolutionary Genetics Analysis*) (Kumar et
148 al. 2018). Phylogenetic analysis was performed under the Maximum Likelihood (ML)
149 method using RAxML-HPC BlackBox available on the CIPRES Science Gateway platform.
150 The GTR + G + I substitution model was applied with 1000 bootstrap replicates to assess
151 the robustness of the phylogenetic nodes (Edler et al. 2021). The obtained sequences
152 were deposited in the GenBank database under the accession numbers LES54
153 (PX954299), LES58 (PX954300), and LES59 (PX954301).

154 **2.5 Biological Cycle**

155 The biological cycle of *Monalonion dissimulatum* was evaluated *in situ*, directly on cacao
156 plants in a productive plot within the study area. For this purpose, rectangular cages
157 measuring 25 × 25 × 30 cm were constructed with a wooden frame and covered with
158 tulle mesh (1 mm aperture), which were installed over selected cacao pods.
159 For the egg stage, 60 eggs deposited endophytically in pod tissues were identified and
160 labeled using markers and adhesive tape to record the incubation period. After hatching,
161 47 viable nymphs were obtained and individually confined within the previously
162 constructed cages.
163 Observations were conducted daily until adult emergence and subsequent natural
164 death, allowing the duration of each developmental stage to be recorded. The evaluated
165 variables included the duration of each biological stage (egg, five nymphal instars, and
166 adult), as well as the total duration of the biological cycle.

167 **2.6 Statistical Analysis**

168 Statistical analyses were performed using R version 4.5.2 (R Core Team 2025). The
169 relationship between incidence and altitude was modeled through Pearson correlation
170 analysis and simple linear regression. The assumptions of normality and homogeneity
171 of residual variances were evaluated using the Shapiro-Wilk and Breusch-Pagan tests,
172 respectively, complemented by Cook's distance for the detection of influential
173 observations. Morphometric variables and biological cycle parameters were
174 summarized using descriptive statistics.

175 **3 RESULTS**

176 In this study, variable incidence levels were recorded across the different evaluated
177 plots. The identification of *Monalonion dissimulatum* was based on a detailed
178 morphological characterization of external structures throughout its biological cycle,
179 establishing diagnostic criteria for its taxonomic delimitation. This morphological
180 evidence was corroborated through molecular analysis, thereby confirming the species-
181 level identification. Finally, the temporal parameters of the biological cycle were
182 determined.

183 **3.1 Incidence and Description of Damage Associated with *Monalonion***

184 Incidence levels ranging from 12 to 75% were recorded across the evaluated plots,
185 showing a positive and statistically significant association with altitude ($r = 0.62$; $p <$
186 0.01). Regression analysis explained 38.5% of the observed variation ($F = 14.4$; $p <$
187 0.01). The slope of the model ($\beta = 0.062$; 95% CI: 0.028–0.096; $p < 0.001$) indicates that
188 for every 100 m increase in altitude, incidence increased by approximately 6.2% (Figure
189 2).

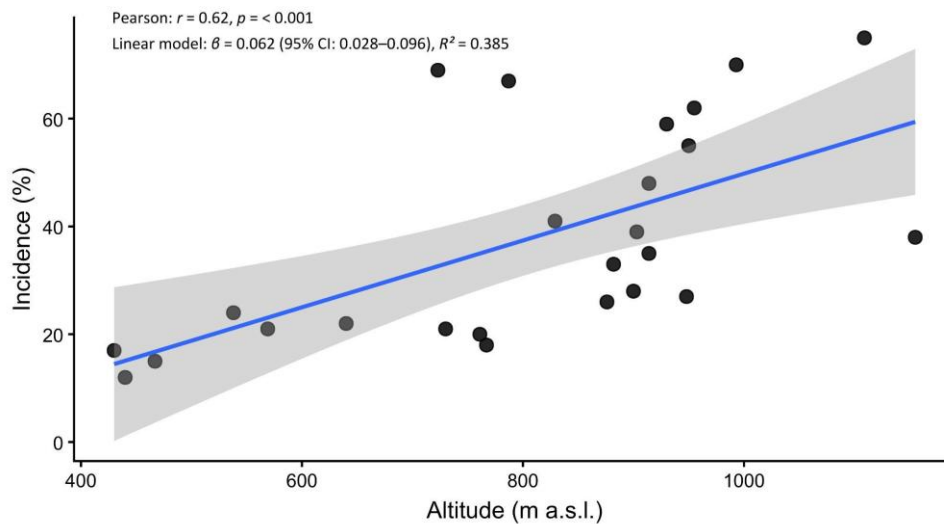


Figure 2: Pearson correlation and linear regression analysis showing the relationship between disease incidence and altitude. The incidence (%) observed in 25 plots is presented as a function of the altitudinal gradient. The solid blue line represents the fitted linear regression model, and the shaded area indicates the 95% confidence interval.

190 Damage was more evident in plots with higher infestation levels. Punctiform necrotic
 191 lesions were observed on developing young fruits, associated with stylet insertion.
 192 These lesions exhibited dark brown to black coloration, accompanied by localized tissue
 193 depression. In some cases, deformation and partial growth arrest were observed,
 194 particularly when the lesions affected a significant portion of the pericarp. Additionally,
 195 signs and symptoms compatible with frosty pod rot of cacao were recorded in infested
 196 fruits (Figure 3).

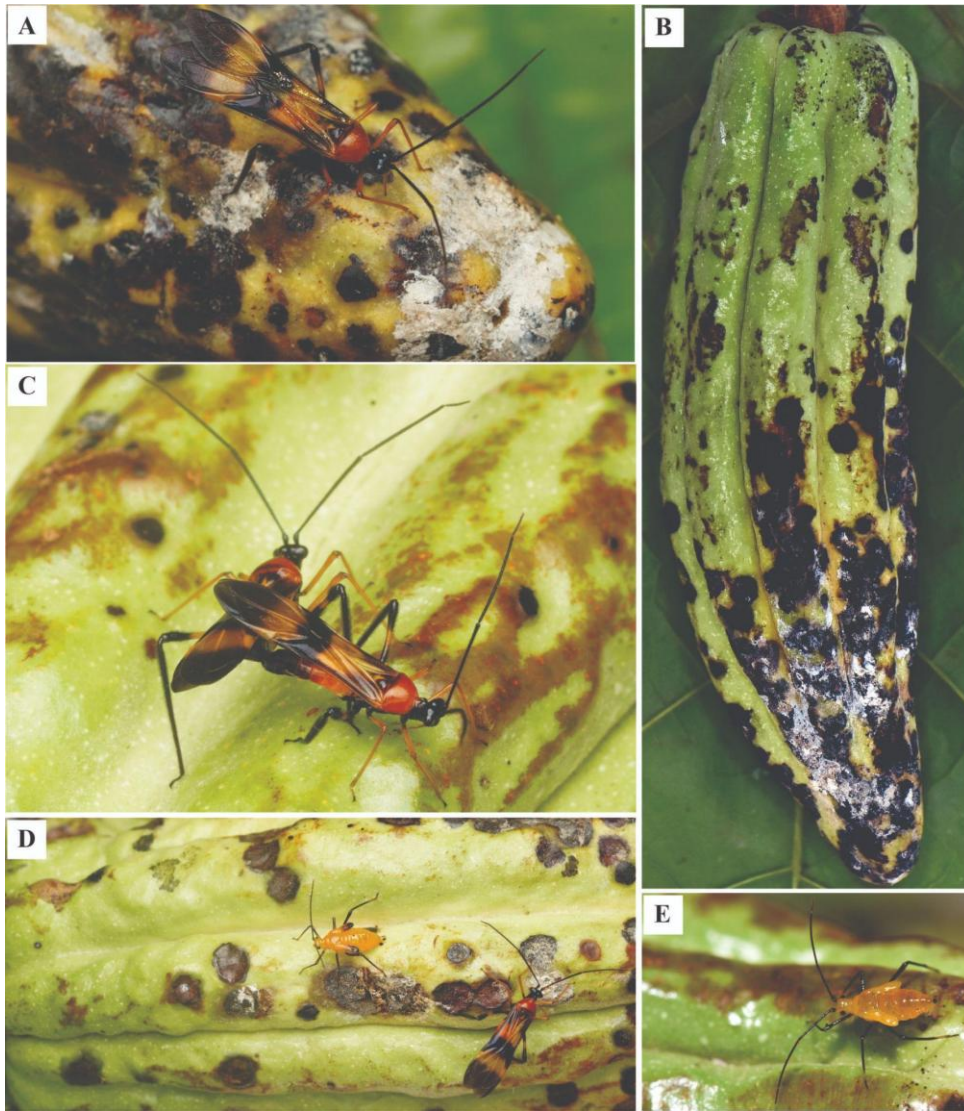


Figure 3: Damage caused by *Monalonion dissimulatum* on cacao fruits. (A) Adult feeding, showing necrotic lesions on the pericarp. (B) Fruit at an early developmental stage exhibiting deformation and the presence of dark necrotic spots, some coalescent, accompanied by mycelium compatible with *Moniliophthora roreri*. (C) Damage caused by the female and male during mating. (D) Adult and nymph associated with circular necrotic lesions accompanied by tissue collapse. (E) Nymph on a young fruit, showing the immature stage responsible for damage through sap-sucking feeding.

197 3.2 Taxonomy

198 ***Monalonion dissimulatum*** Distant, W. L., 1883. *Biologia Centrali-Americana. Insecta.*
199 *Rhynchota. Hemiptera-Heteroptera*, Vol. 1. Taylor & Francis, London.

200 *Monalonion dissimulatum*, belonging to the family Miridae and the tribe Monaloniini,
201 exhibits paurometabolous development comprising an egg stage and five nymphal
202 instars before reaching the adult stage. The eggs, oviposited endophytically within cacao
203 pods, possess two filiform appendages or aeropyles that protrude from the plant tissue
204 and remain in contact with the external environment, allowing gaseous exchange during
205 embryonic development. These ovigerous structures are characterized by a milky-white
206 coloration, an elongated shape, and a slight curvature.

207 During nymphal instars I and II, the nymphs of *M. dissimulatum* exhibit a bright, light
208 yellow-orange coloration and limited mobility on the surface of the cacao pod. In the
209 first nymphal instar, an average body length of 2.09 mm and a body width of 0.56 mm
210 were recorded, whereas in the second nymphal instar the mean body length was 2.55
211 mm and the body width reached approximately 0.87 mm. In both stages, feeding activity
212 was reduced and largely restricted to areas near the hatching site.

213 The third nymphal instar was characterized by a notable increase in mobility and
214 feeding activity on the cacao pod. Nymphs exhibited a bright orange coloration, and
215 developing wing primordia became distinguishable. At this stage, the average body
216 length was 3.30 mm and the body width was 1.58 mm. Nymphs corresponding to the
217 fourth instar maintained a bright yellow-orange coloration and displayed more
218 developed and conspicuous wing primordia compared with the previous stages. In this
219 phase, the average body length reached 4.28 mm, with a body width of approximately
220 1.75 mm, reflecting a greater overall robustness of the insect.

221 In the fifth nymphal instar, nymphs exhibited a coloration similar to that of the fourth
222 instar; however, the wing primordia developed a darker pigmentation, particularly
223 toward the apex. As the molting process to the adult stage approached, a pronounced
224 elongation of the abdomen was observed. At this stage, the average body length was
225 9.63 mm and the body width reached approximately 2.78 mm, representing the largest
226 and most morphologically developed nymphal phase.

227 The morphological characterization of adults was performed through the analysis of
228 each body segment.

229 **Head.** Shiny black in color and of the prognathous type, with a smooth and strongly
230 sclerotized surface. The compound eyes are large, prominent, and laterally positioned.
231 In lateral view, the cephalic capsule exhibits a well-defined contour and a marked
232 separation between the frontal region and the clypeus. On the ventral region of the head,
233 the proboscis characteristic of the piercing sucking mouthparts is articulated. This
234 structure is elongated and straight and is composed of four well-differentiated
235 segments. The first segment is darkly colored; the second segment displays a bicolored
236 pattern, with the basal portion dark and the distal portion yellowish; whereas the third
237 and fourth segments are predominantly yellowish, progressively tapering toward the
238 apex.

239 **Antennae.** Filiform, long, and slender, exceeding the length of the head and pronotum.
 240 They are composed of four antennomeres with predominantly black coloration. The
 241 surface of the antennomeres is covered with dark setae. The fourth antennomere
 242 exhibits a yellowish coloration on its dorsal portion.

243 **Thorax.** Differentiated into the prothorax, mesothorax, and metathorax. The prothorax
 244 presents, in its anterior portion, a narrow region with black coloration and a cervical-
 245 like appearance, with a width smaller than that of the head. Posteriorly, the pronotum
 246 gradually widens and extends ventrally on both sides of the body. In the dorsal view, the
 247 pronotum is bell-shaped and exhibits an intense yellow-orange coloration; the posterior
 248 margin shows a dark spot in its median region that extends backward and terminates
 249 in a pointed projection. In the ventral view, the lower left and right margins of the
 250 pronotum also display dark pigmentation.

251 On the ventral region of the prothorax, the forelegs are articulated and are of the
 252 cursorial type. These appendages bear abundant yellow-orange setae. The coxa is dark
 253 on its external surface and caramel-colored on the internal surface, whereas the
 254 trochanter is yellowish. The femur is predominantly yellow, with a slight dark spot at
 255 its base near the articulation with the trochanter. The tibia is yellow, as are the first two
 256 tarsal segments and the basal half of the third segment; the distal half of the last tarsal
 257 segment and the claws (unguis) are dark in color.

258 In the anterodorsomedial region of the mesothorax, the scutellum is clearly
 259 distinguishable, exhibiting a triangular shape. It presents two dark lateral spots, one on
 260 each side, shaped like inverted triangles, separated by a central yellow spot that is also
 261 triangular in form. The mesosternum, in its anterior portion and anterior to the coxae,
 262 displays a central band and lateral margins of black coloration, which delimit on both
 263 sides two oval caramel-colored spots.

264 Within the mesothorax, the middle legs also of the cursorial type are articulated. The
 265 coxa is dark on its external surface and caramel-colored on the internal surface, while
 266 the trochanter is dark. The femur is yellow, with dark pigmentation restricted to its
 267 basal or proximal portion (approximately one-sixth of its length). The tibia and the first
 268 two tarsal segments are yellow; the distal half of the third tarsal segment and the claws
 269 are dark in color.

270 Dorsally, the mesothorax bears the first pair of wings, or hemelytra. These structures
 271 exhibit a general yellow-orange coloration. On the corium, a dark spot is observed in the
 272 basal portion, which progressively narrows parallel to the costal margin and extends
 273 toward the posterior region, delimiting a yellowish spot in the central area. The
 274 membrane presents a basal dark spot followed by a yellow band; from the midpoint
 275 toward the distal end, a well-defined dark band is observed.

276 On the ventral region of the metathorax, the hind legs are longer and also of the cursorial
 277 type are articulated. The coxa and trochanter are dark in color. The femur exhibits dark
 278 pigmentation on its basal and distal thirds, while the middle third is yellow. The tibia
 279 and tarsus are predominantly dark, and all segments are provided with dark setae. On
 280 the dorsal region of the metathorax, the hind wings are articulated; these are
 281 membranous, yellow-orange in coloration, and bear two transverse dark spots: the first

282 narrowed in its median region and the second located at the distal or apical end of the
 283 wing.
 284 **Abdomen.** Elongated and segmented, consisting of nine visible segments in females and
 285 eight in males. The first two abdominal segments exhibit a yellow–orange coloration,
 286 which contrasts markedly with the posterior segments that show predominantly dark
 287 pigmentation intensifying toward the distal end. In the dorsal view, the abdomen is
 288 partially covered by the hemelytra and hind wings, leaving only the terminal segments
 289 exposed. In ventral and lateral views, the abdominal segments display a progressive
 290 chromatic differentiation, with intersegmental areas of lighter and more membranous
 291 appearance. The terminal segment is dark in color and bears short, scattered setae,
 292 more evident at the apex where the external genital structures are located.
 293 The descriptions are detailed in **Figure 4**.



Figure 4: Morphological characters of *Monalonion dissimulatum*. (A) Live adult in lateral view. (B) Lateral view of the head. (C) Dorsal view of the head. (D) Antenna

showing four visible segments. (E) Detail of the junction between the antennal scape and pedicel. (F) Terminal antennal segment of the male. (G) Terminal antennal segment of the female. (H) Rostrum with four segments. (I) Dorsal view of the pronotum and posterior margin of the scutellum. (J) Dorsal view of the scutellum. (K) Dorsal view of the hemelytra. (L) Legs showing the characteristic coloration and the yellow band on the median portion of the hind femora. (M) Forewings (upper image) and hind wings (lower image), showing the coloration pattern. (N) Female terminalia.

294 Following detailed morphological analysis, differences were detected in the
295 conformation of the abdominal segments and in body size between males and females;
296 however, sexual dimorphism is not readily evident to the naked eye (Figure 3C). The
297 corresponding morphometric measurements are presented in **Table 1**.

Table 1. Mean morphometric measurements (mm) of male and female specimens (n = 45 per sex).

Measurement (mm)	Male	Female
Body		
Length	11.36	12.92
Width	2.23	2.37
Head		
Length	0.91	0.99
Width	1.49	1.71
Vertex width	0.63	0.63
Antenna		
Segment I	0.44	0.52
Segment II	4.22	4.81
Segment III	4.02	4.56
Segment IV	1.18	1.19
Pronotum		
Length	1.38	1.5
Basal width	0.91	0.99
Hemelytron		
Length	9.15	11.54
Cuneus		
Length	2.09	2.55
Basal width	0.44	0.54
Legs		
Fore femur	2.23	2.7
Middle femur	2.41	2.68
Hind femur	3.58	3.91
Fore tibia	3.35	3.69
Middle tibia	3.6	3.84
Hind tibia	5.28	5.91

298 3.3 Molecular Identification

299 Taxonomic identity was confirmed through comparisons in the BOLD Systems database,
 300 where the obtained sequences showed 100% similarity with the reference sequence
 301 (TRBBO262-11) previously reported, corroborating species-level identification.
 302 Additionally, phylogenetic analysis based on mitochondrial COI gene sequences was
 303 performed using the Maximum Likelihood method. The results showed that the
 304 sequences obtained from specimens LES54, LES58, and LES59 consistently clustered
 305 with the reference sequence of *Monalonion dissimulatum*, revealing their close
 306 phylogenetic relationship and confirming the species identification (Figure 5).
 307

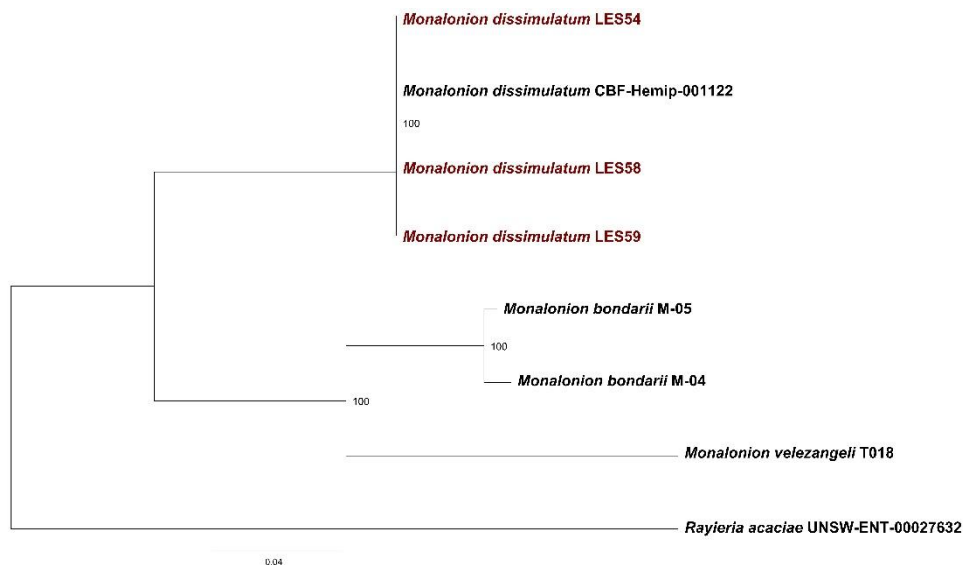


Figure 5: Phylogenetic tree based on the COI gene, constructed using the Maximum Likelihood method (RAxML) with 1000 bootstrap replicates. The sequences LES54, LES58, and LES59 (in red) cluster with the reference TRBBO262-11 (*Monalonion dissimulatum*), while *Rayieria acaciae* was used as the outgroup.

308 3.4 Biological Cycle

309 The complete biological cycle of *M. dissimulatum* under field conditions lasted between
 310 34 and 46 days and comprised seven developmental stages. The egg incubation period
 311 ranged from 6 to 10 days, whereas the complete nymphal development (five instars)
 312 required between 20 and 31 days. Adult longevity ranged from 8 to 10 days (**Table 2**).
 313

314 Table 2. Duration (days) of the biological cycle of *M. dissimulatum*.

Developmental stage	Duration (days)
Egg	6–10
Nymph I	4–6
Nymph II	4–5
Nymph III	4–5
Nymph IV	4–5
Nymph V	4–5
Adult	8–10
Total life cycle	34–46

315 Values represent the observed range (minimum–maximum).

316 4 DISCUSSION

317 In the Neotropics, the genus *Monalonion* comprises polyphagous mirids of economic
 318 importance (Gamboa et al. 2020). In *Theobroma cacao*, the species *M. bondari* and *M.*
 319 *dissimulatum* have been identified as phytophagous insects that damage both vegetative
 320 and reproductive organs (Gamboa et al. 2020; Chirinos et al. 2025). Nevertheless,
 321 taxonomic and ecological knowledge of these taxa within cacao agroecosystems
 322 remains limited.

323 Given the limitations of morphological characters for species discrimination (Gamboa
 324 et al. 2020), his study validated the identity of *M. dissimulatum* through an integrative
 325 taxonomic approach combining morphological and molecular analyses.
 326 Morphologically, sexual dimorphism was subtle and mainly restricted to body size.
 327 Previous studies have indicated that coloration in adults may vary between sexes;
 328 however, although such variation occurs, this attribute does not constitute a reliable
 329 diagnostic character for species delimitation due to its high variability (Gamboa et
 330 al. 2020). At the molecular level, the COI gene corroborated the taxonomic identity
 331 through phylogenetic proximity with reference sequences, reinforcing its effectiveness
 332 as a marker for insect identification (Folmer et al. 1994; Hebert et al. 2003).

333 Specifically, there are no previous records regarding the altitudinal range of *M.*
 334 *dissimulatum*, although other species of the genus *Monalonion* have been reported to
 335 occur between 600 and 1800 m a.s.l. (Chirinos et al. 2025). In the present study, the pest
 336 exhibited a wide distribution across cacao plantations, spanning an altitudinal range
 337 from 430 to 1155 m a.s.l. Incidence showed a positive correlation with altitude;
 338 however, this factor explained only 38% of the observed variance. In addition to
 339 altitude, abiotic factors (temperature, relative humidity, and microclimatic conditions)
 340 and biotic factors (host availability and plot composition) may influence the population
 341 dynamics and behavior of the species along the altitudinal gradient (Andrew et al. 2012;
 342 Dostálek et al. 2018; Paudel et al. 2021). Consequently, the higher incidence recorded
 343 at greater altitudes could be associated with environmental conditions that favor the
 344 development and survival of *M. dissimulatum*.

345 Discrete necrotic lesions and tissue deformations recorded in young fruits correspond
 346 to the typical damage caused by piercing-sucking insects, confirming the observations

347 reported by (Vargas et al. 2005). The feeding mechanism, based on stylet insertion,
 348 induces tissue necrosis that alters fruit morphology (Babin 2018). Studies conducted in
 349 cacao and coffee crops have corroborated that such lesions may evolve into dark spots
 350 that compromise the integrity of the pericarp (Giraldo-Jaramillo et al. 2010; Valarezo
 351 Cely et al. 2012). Critically, these discontinuities in the epidermis facilitate secondary
 352 infection by opportunistic pathogen (Bailey et al. 2018; Gossner et al. 2021),
 353 particularly frosty pod rot (*Moniliophthora roreri*), one of the most destructive diseases
 354 affecting cacao (Díaz-Valderrama et al. 2020; Leiva et al. 2022; Santillan-Culquimboz et
 355 al. 2025).

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 359 induces tissue necrosis that alters fruit morphology (Babin, 2018). Studies conducted in
 360 cacao and coffee crops have corroborated that such lesions may evolve into dark spots
 361 that compromise the integrity of the pericarp (Giraldo-Jaramillo et al., 2010; Valarezo
 362 Cely et al., 2012). Critically, these discontinuities in the epidermis facilitate secondary
 363 infection by opportunistic pathogens (Bailey et al., 2018; Gossner et al., 2021),
 364 particularly frosty pod rot (*Moniliophthora roreri*), one of the most destructive diseases
 365 affecting cacao (Díaz-Valderrama et al., 2020; Leiva et al., 2022; Santillan-Culquimboz
 366 et al., 2025).

367 The biological cycle of *M. dissimulatum* under field conditions ranged from 34 to 46 days,
 368 encompassing five nymphal instars before reaching adulthood, consistent with previous
 369 reports (Villacorta 1977; Giraldo-Jaramillo et al. 2010; Flórez et al. 2018). These values
 370 are consistent with the biology of the genus *Monalonion*, which is characterized by short
 371 life cycles and high reproductive rates in tropical environments (Gamboa et al. 2024).
 372 The brevity of the cycle suggests a high potential for population increase, facilitating the
 373 rapid colonization of new plots under favorable environmental conditions.

374 From an ecological and phytosanitary perspective, the presence of *M. dissimulatum* in
 375 cacao plantations constitutes a threat to the production system, given its status as a key
 376 pest capable of compromising fruit yield and quality when population levels increase
 377 (Castillo 2013; Cuellar-Palacios et al. 2024). The combination of a short biological cycle,
 378 the mobility of both nymphal and adult stages, its polyphagous feeding behavior, and its
 379 role as a disease vector increases its potential as a pest of economic importance, thereby
 380 demanding comprehensive monitoring and control strategies (Vargas Serna et al. 2020;
 381 Iñiguez Freitas 2025).

382 5 CONCLUSIONS

383 The incidence of *Monalonion dissimulatum* in the evaluated cacao plots showed a
 384 positive and significant correlation with altitude, with infestation levels reaching up to
 385 75% in the highest altitudinal gradients. Direct damage, characterized by necrotic
 386 lesions and deformation of the pericarp, together with the evidence of association with
 387 mycelium of frosty pod rot, confirms the role of *M. dissimulatum* as a key pest and a
 388 potential vector of diseases. The detailed morphological characterization of the

developmental stages, although useful for diagnostic description, was insufficient to distinguish sexual dimorphism by simple visual inspection. Confirmation using the molecular marker COI corroborated the species identity. Biologically, the life cycle under field conditions was completed in 34 to 46 days, encompassing five nymphal instars. Overall, these findings establish essential biological and ecological foundations for improving species identification and optimizing Integrated Pest Management strategies in cacao systems.

Author Contributions

Conceptualization: S.T.L.E., L.A.C.V., J.A.V.C.; Data curation: S.T.L.E., H.W.S.C., V.A.R., J.A.V.C.; Formal analysis: S.T.L.E., H.W.S.C., V.A.R., J.A.V.C.; Funding acquisition: S.T.L.E., G.F.R.D., J.A.V.C.; Investigation: S.T.L.E., H.W.S.C., G.F.R.D., D.B.M., V.A.R., E.R.R.R., L.A.C.V., J.A.V.C.; Methodology: S.T.L.E., H.W.S.C., L.A.C.V., J.A.V.C.; Project administration: S.T.L.E., L.A.C.V.; Resources: S.T.L.E., H.W.S.C., G.F.R.D., D.B.M., V.A.R., E.R.R.R., L.A.C.V., J.A.V.C.; Software: S.T.L.E., V.A.R., J.A.V.C.; Supervision: S.T.L.E., H.W.S.C., G.F.R.D., V.A.R., L.A.C.V., J.A.V.C.; Validation: S.T.L.E., H.W.S.C., G.F.R.D., D.B.M., V.A.R., E.R.R.R., L.A.C.V., J.A.V.C.; Visualization: S.T.L.E., H.W.S.C., D.B.M., V.A.R., E.R.R.R., L.A.C.V.; Writing – original draft: S.T.L.E., H.W.S.C., G.F.R.D., D.B.M., V.A.R., E.R.R.R., L.A.C.V., J.A.V.C.; Writing – review and editing: S.T.L.E., H.W.S.C., G.F.R.D., D.B.M., V.A.R., E.R.R.R., L.A.C.V., J.A.V.C. All authors approved the final version of the manuscript.

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Conflicts of interest

The authors declare no conflict of interest.

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